

CKD-MBD CASCADE

SECONDARY HPT
Ca DOWN
PTH UP

TERTIARY HPT
Ca UP
PTH UP

EARLY CKD
STEP 1: GFR DOWN
PO₄ RETAINED

OSTEOCYTE ALARM
STEP 2: FGF23 UP

1α-HYDROXYLASE
STEP 3: CALCITRIOL DOWN

GUT Ca DOWN
STEP 4

SECONDARY HPT
STEP 5: PTH UP
SECONDARY HPT LAB
APPROPRIATE RESPONSE
Ca DOWN/NORMAL
PO₄ UP
1,25-D DOWN
PTH UP

HIGH-TURNOVER BONE
BONE PAIN
FRACTURE
OSTEITIS FIBROSA
DON'T OVER-SUPPRESS PTH

ADYNAMIC / LOW-TURNOVER BONE
OVER-SUPPRESSED PTH
Ca OVERLOAD
DM / ELDERLY

TERTIARY HPT
AUTONOMOUS
POST-TRANSPLANT
PTH UP UP
REFRACTORY
PARATHYROIDECTOMY

MEDICAL MANAGEMENT
DIETARY PO₄ RESTRICTION
SEVELAMER (PHOSPHATE BINDER, NO Ca LOAD)
Ca-BASED BINDER (Ca LOAD)
VIT D ANALOG
CINACALCET / CALCIMIMETIC (CaSR-key, ↓ PTH)
AVOID Ca/PO₄ EXCESS

1. Step 1: GFR down / PO4 retained

WHAT YOU SEE

Failing kidney over basin: 'STEP 1 — GFR DOWN / PO4 RETAINED, EARLY CKD'.

WHAT IT ENCODES

The cascade begins as GFR falls: the kidney can no longer excrete the daily phosphate load, so phosphate is retained — the first trigger of CKD-MBD. Phosphate retention typically begins around CKD stage 3 (eGFR <60), though serum phosphate may stay normal for a while because FGF23 and PTH ramp up phosphaturia to compensate. Hyperphosphatemia directly lowers calcium (binds it) and stimulates PTH.

BOARD TRAP

Reassurance from a normal serum phosphate early in CKD is the trap — phosphate retention starts before the level rises overtly, masked by rising FGF23/PTH, so don't wait for hyperphosphatemia to begin counseling on phosphate control.

2. Step 2: FGF23 / osteocyte alarm

WHAT YOU SEE

Bone cells throwing 'FGF23' alarms: 'STEP 2 — OSTEOCYTE ALARM'.

WHAT IT ENCODES

Osteocytes sense the phosphate load and release FGF23 (the earliest hormone to rise in CKD-MBD). FGF23 promotes renal phosphate excretion (phosphaturia) — helpful — but it also SUPPRESSES 1-alpha-hydroxylase, cutting calcitriol production, which is harmful. So the body's first compensatory move simultaneously sets up the calcitriol deficiency that drives secondary hyperparathyroidism.

BOARD TRAP

Thinking FGF23 is purely protective is the trap — by suppressing calcitriol it lowers gut calcium absorption and accelerates secondary HPT; FGF23 is the earliest measurable abnormality, rising before phosphate or PTH are overtly off.

3. Step 3: calcitriol down (1 α -hydroxylase blocked)

WHAT YOU SEE

A ninja blocking '1 α -HYDROXYLASE': 'STEP 3 — CALCITRIOL DOWN'.

WHAT IT ENCODES

Reduced functional renal mass plus FGF23 suppression cut 1-alpha-hydroxylase activity, so the kidney can't convert 25-OH vitamin D to active 1,25-(OH)₂ vitamin D (calcitriol). Low calcitriol means less intestinal calcium absorption and loss of calcitriol's direct suppression of PTH gene transcription — both push PTH up. This step is why CKD patients need calcitriol or analogs (paricalcitol, doxercalciferol), not plain vitamin D.

BOARD TRAP

Prescribing cholecalciferol/ergocalciferol to correct the 'vitamin D deficiency' of CKD is the trap — the diseased kidney cannot 1-alpha-hydroxylate it, so you must give ACTIVE vitamin D (calcitriol or an analog) that bypasses the blocked enzyme.

4. Step 4: gut Ca absorption down

WHAT YOU SEE

A sad gut: 'STEP 4 — GUT Ca DOWN'.

WHAT IT ENCODES

Low calcitriol reduces intestinal calcium absorption (calcitriol normally upregulates the gut calcium transporters), producing hypocalcemia. The hypocalcemia, hyperphosphatemia, and loss of calcitriol's direct suppression all converge on the parathyroid to drive PTH up. This is the link between the vitamin D arm and the calcium/PTH arm of the cascade.

BOARD TRAP

Treating the low calcium of CKD with calcium-based binders or high-dose calcium is tempting but is the trap — it can push the Ca $\times$ PO₄ product up and worsen vascular calcification; correct the underlying calcitriol deficiency instead.

5. Step 5: secondary HPT (PTH up)

WHAT YOU SEE

An enlarged gland: 'STEP 5 — SECONDARY HPT, PTH UP, PO₄ UP, Ca down, calcitriol DOWN'.

WHAT IT ENCODES

Three stimuli — hypocalcemia, hyperphosphatemia, and low calcitriol — chronically drive the parathyroids, producing secondary hyperparathyroidism: HIGH PTH with LOW or normal calcium. Over time the glands hyperplasiae. The defining lab signature is high PTH + high PO₄ + low/normal Ca + low calcitriol, distinguishing it from the high-Ca hyperparathyroid states.

BOARD TRAP

Confusing the three hyperparathyroid states is the high-yield trap: PRIMARY HPT = high Ca + high PTH (low PO₄); SECONDARY HPT (CKD) = LOW/normal Ca + high PTH (high PO₄); TERTIARY HPT = HIGH Ca + high PTH (autonomous). The calcium direction sorts them.

6. Secondary HPT lab box

WHAT YOU SEE

A lab-results card on the bench topped with a rack of blood-test vials, labelled 'SECONDARY HPT LAB' — the vials are the biochemical fingerprint (high PTH, high PO₄, low/normal Ca, low calcitriol) that confirms the cascade.

WHAT IT ENCODES

Classic secondary HPT panel in CKD: HIGH phosphate, HIGH PTH, LOW/normal calcium, LOW calcitriol, and elevated FGF23. KDIGO targets in CKD aim to lower phosphate toward normal, keep calcium in range, and keep PTH within ~2–9x the upper limit of normal (NOT fully normal) on dialysis.

BOARD TRAP

Reading a 'low-normal calcium with markedly high PTH' as primary hyperparathyroidism is the trap — in primary HPT calcium is HIGH; the low/normal calcium with high PTH and high phosphate plus impaired renal function is secondary HPT of CKD.

7. High-turnover bone / osteitis fibrosa

WHAT YOU SEE

A painful bone: 'HIGH-TURNOVER BONE — BONE PAIN, FRACTURE, OSTEITIS FIBROSA'.

WHAT IT ENCODES

Persistently high PTH drives high-turnover renal osteodystrophy — osteitis fibrosa cystica — with increased osteoclastic resorption, marrow fibrosis, bone pain, and fractures. This is the under-treated end of the spectrum. Histology shows elevated bone turnover; treatment is controlling phosphate, repleting calcitriol, and using calcimimetics to lower PTH.

BOARD TRAP

Assuming any CKD bone disease is high-turnover is the trap — over-correcting PTH flips the patient to the OPPOSITE problem, low-turnover adynamic bone, which is now common because of aggressive PTH suppression.

8. Don't over-suppress PTH / adynamic bone

WHAT YOU SEE

Box: 'DON'T OVER-SUPPRESS PTH; ADYNAMIC/LOW-TURNOVER BONE — Ca OVERLOAD, ELDERLY'.

WHAT IT ENCODES

If therapy drives PTH TOO low (over-suppression with calcium, calcitriol, calcimimetics), bone turnover halts — adynamic (low-turnover) bone disease. The 'frozen' bone can't buffer a calcium load, so calcium spills into vessels → hypercalcemia, vascular calcification, and fractures. Risk is higher in elderly and diabetic dialysis patients.

BOARD TRAP

Normalizing PTH all the way to the standard reference range in a CKD patient is the trap — some elevation of PTH is needed to maintain bone turnover, so KDIGO targets a PTH 2–9x ULN; over-suppression causes adynamic bone.

9. Tertiary HPT (autonomous, post-transplant)

WHAT YOU SEE

Multiple enlarged glands: 'TERTIARY HPT — AUTONOMOUS, POST-TRANSPLANT, PTH UP, Ca UP, REFRACTORY'.

WHAT IT ENCODES

After years of secondary HPT, the chronically stimulated parathyroid glands become AUTONOMOUS (nodular hyperplasia → tertiary HPT): they secrete PTH independent of calcium, so calcium now rises. Often unmasked after renal transplant when the kidney works again but the glands keep over-secreting. Labs: HIGH PTH and HIGH calcium, frequently refractory to medical therapy.

BOARD TRAP

The discriminator from secondary HPT is calcium: tertiary shows HIGH calcium (autonomous secretion), whereas secondary shows low/normal calcium. A CKD/post-transplant patient who becomes HYPERcalcemic with high PTH has progressed to tertiary HPT.

10. Parathyroidectomy (refractory tertiary)

WHAT YOU SEE

A surgeon: 'PARATHYROIDECTOMY'.

WHAT IT ENCODES

Surgical parathyroidectomy (often subtotal or total with autotransplantation) is indicated for refractory tertiary HPT, or severe secondary HPT that fails medical therapy (uncontrollable hyperphosphatemia/PTH, calciphylaxis, intractable pruritus/bone pain). Watch for hungry bone syndrome — profound post-op hypocalcemia as demineralized bone rapidly reuptakes calcium.

BOARD TRAP

Forgetting hungry bone syndrome after parathyroidectomy is the trap — the suddenly PTH-deprived, demineralized skeleton avidly takes up calcium, causing severe prolonged post-op hypocalcemia that needs aggressive calcium and calcitriol.

11. Dietary PO4 restriction

WHAT YOU SEE

Management shelf: 'DIETARY PO4 RESTRICTION'.

WHAT IT ENCODES

First-line management of CKD-MBD is dietary PHOSPHATE restriction (limit dairy, cola, processed foods/preservatives, which are high in absorbable inorganic phosphate). Controlling phosphate is the foundation because hyperphosphatemia drives the whole cascade (FGF23, low calcitriol, secondary HPT, vascular calcification).

BOARD TRAP

Overlooking dietary phosphate while reaching straight for binders/calcitriol is the trap — diet (especially hidden phosphate additives in processed food) is the first and cheapest intervention; binders are added when diet alone is insufficient.

12. Sevelamer (non-Ca binder)

WHAT YOU SEE

A pill bottle on the 'MEDICAL MANAGEMENT' shelf labelled 'SEVELAMER' — the NON-calcium phosphate binder, set apart from the calcium-based bottle to flag it as the calcium-free option for hypercalcemia/vascular calcification.

WHAT IT ENCODES

Sevelamer and lanthanum carbonate are NON-calcium phosphate binders taken WITH meals to block intestinal phosphate absorption without adding a calcium load. Preferred when the patient is hypercalcemic, has vascular calcification, or has adynamic bone. Sevelamer also modestly lowers LDL cholesterol.

BOARD TRAP

Defaulting to calcium-based binders in a patient with hypercalcemia or vascular calcification is the trap — non-calcium binders (sevelamer, lanthanum) avoid the added calcium load that worsens calcification and adynamic bone.

13. Ca-based binder

WHAT YOU SEE

A pill bottle on the management shelf labelled 'Ca-BASED BINDER' — calcium acetate/carbonate that both binds gut phosphate AND adds a calcium load, the trade-off the bottle represents.

WHAT IT ENCODES

Calcium acetate and calcium carbonate are inexpensive phosphate binders that double as a calcium source, useful when the patient is hypocalcemic. But they add a calcium load — limited by hypercalcemia and the risk of vascular calcification, especially with concurrent active vitamin D or in adynamic bone.

BOARD TRAP

Using calcium-based binders in a patient who is already hypercalcemic or has low-turnover/adynamic bone is the trap — the calcium load worsens hypercalcemia and vascular calcification; switch to a non-calcium binder in those settings.

14. Cinacalcet / calcimimetic + avoid $\text{Ca} \times \text{PO}_4$ excess

WHAT YOU SEE

Box: 'CINACALCET / CALCIMIMETIC; AVOID $\text{Ca} \times \text{PO}_4$ EXCESS', plus 'VIT D ANALOG'.

WHAT IT ENCODES

Calcimimetics (cinacalcet, IV etelcalcetide) allosterically sensitize the parathyroid CaSR to calcium, lowering PTH AND calcium — useful when PTH is high but calcium is also high. Active vitamin D analogs (calcitriol, paricalcitol) lower PTH but raise calcium and phosphate. Keep the $\text{Ca} \times \text{PO}_4$ product low ($< \sim 55$) to limit vascular calcification.

BOARD TRAP

Choosing active vitamin D to lower PTH in a patient with high calcium or high $\text{Ca} \times \text{PO}_4$ product is the trap — vitamin D analogs raise both, worsening calcification; cinacalcet lowers PTH while also LOWERING calcium, making it the right tool when calcium is elevated.
